ÁLGEBRA LINEAL

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CARRERA DE PEDAGOGÍA EN LAS CIENCIAS EXPERIMENTALES
MATEMÁTICA Y FÍSICA

Tarea Individual 06

-Matriz Inversa y Matrices Elementales-

Actividad: Escribir la inversa de la matriz A como producto de matrices elementales.

$$A = \begin{pmatrix} 2 & 4 & 6 \\ 4 & 5 & 6 \\ 3 & 1 & -2 \end{pmatrix} \tag{1}$$

1. **Calculamos** A^-1

$$\left(\begin{array}{ccc|ccc} 2 & 4 & 6 & 1 & 0 & 0 \\ 4 & 5 & 6 & 0 & 1 & 0 \\ 3 & 1 & -2 & 0 & 0 & 1 \end{array}\right) \xrightarrow{F_1/2 \to F_1} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1/2 & 0 & 0 \\ 4 & 5 & 6 & 0 & 1 & 0 \\ 3 & 1 & -2 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[-3F_1+F_3 \to F_3]{-4F_1+F_2 \to F_2} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1/2 & 0 & 0 \\ 0 & -3 & -6 & -2 & 1 & 0 \\ 0 & -5 & -11 & -3/2 & 0 & 1 \end{array}\right) \xrightarrow{F_2/-3 \to F_3} \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1/2 & 0 & 0 \\ 0 & 1 & 2 & 2/3 & -1/3 & 0 \\ 0 & -5 & -11 & -3/2 & 0 & 1 \end{array}\right)$$

$$\xrightarrow[5F_2+F_3 \to F_3]{-2F_2+F_1 \to F_1} \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & -5/6 & 2/3 & 0 \\ 0 & 1 & 2 & 2/3 & -1/3 & 0 \\ 0 & 0 & -1 & 11/6 & -5/3 & 1 \end{array}\right) \xrightarrow{-F_3 \to F_3} \left(\begin{array}{ccc|ccc} 1 & 0 & -1 & -5/6 & 2/3 & 0 \\ 0 & 1 & 2 & 2/3 & -1/3 & 0 \\ 0 & 0 & 1 & -11/6 & 5/3 & -1 \end{array}\right)$$

$$\xrightarrow[F_3+F_1 \to F_1]{-2F_3+F_2 \to F_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -8/3 & 7/3 & -1 \\ 0 & 1 & 0 & 13/3 & -11/3 & 2 \\ 0 & 0 & 1 & -11/6 & 5/3 & -1 \end{array}\right)$$

En conlusión:

$$\mathrm{A}^-1 = \begin{pmatrix} -8/3 & 7/3 & -1 \\ 13/3 & -11/3 & 2 \\ -11/6 & 5/3 & -1 \end{pmatrix} \tag{2}$$

2. **Expresa como producto de matrices elementales**

$$B = E_1 A = \begin{pmatrix} -3/8 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -8/3 & 7/3 & -1 \\ 13/3 & -11/3 & 2 \\ -11/6 & 5/3 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -7/8 & 3/8 \\ 13/3 & -11/3 & 2 \\ -11/6 & 5/3 & -1 \end{pmatrix}$$

$$C = E_2 B = \begin{pmatrix} 1 & 0 & 0 \\ -13/3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -7/8 & 3/8 \\ 13/3 & -11/3 & 2 \\ -11/6 & 5/3 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1/8 & 3/8 \\ -11/6 & 5/3 & -1 \end{pmatrix}$$

$$D = E_3 C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 11/6 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1/8 & 3/8 \\ -11/6 & 5/3 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1/8 & 3/8 \\ 0 & 1/16 & -5/16 \end{pmatrix}$$

$$E = E_4D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1/8 & 3/8 \\ 0 & 1/16 & -5/16 \end{pmatrix} = \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1 & 3 \\ 0 & 1/16 & -5/16 \end{pmatrix}$$

$$F = E_5E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1/16 & 1 \end{pmatrix} \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1 & 3 \\ 0 & 1/16 & -5/16 \end{pmatrix} = \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1 & 3 \\ 0 & 0 & -1/2 \end{pmatrix}$$

$$G = E_6F = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1 & 3 \\ 0 & 0 & -1/2 \end{pmatrix} = \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$$

$$H = E_7G = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$I = E_8H = \begin{pmatrix} 1 & 7/8 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -7/8 & 3/8 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 3/8 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$I = E_8H = \begin{pmatrix} 1 & 0 & -3/8 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3/8 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\boxed{\therefore E_9E_8E_7E_6E_5E_4E_3E_2E_1 = I}$$

Referencias Bibliógraficas

Grossman, S. (2012). *Álgebra Lineal* 7ed. McGRAW–HILL\INTERAMERICANA EDITORES, S.A. DE C.V.